Chromatic number does not determine configuration-space Betti numbers

A counterexample to TLMC conjecture 00000004091

14 October 2026

Abstract

The polynomial clause of conjecture 00000004091 is false already for one particle. The graphs K_2 and C_4 both have chromatic number 2, whereas $\mathrm{Conf}_1(K_2)$ and $\mathrm{Conf}_1(C_4)$ have first Betti numbers 0 and 1. We give the complete elementary argument and a self-contained Lean 4.31.0 verification of the color-minimality and rational cellular-homology certificates. The subdivision-invariance clause is not challenged.

1 Statement and interpretation

The source conjecture [1] concerns the unordered configuration space of n pairwise distinct points of a graph. It asserts both subdivision invariance of the first Betti number and its expression as a polynomial in the graph's chromatic number, of degree exactly n. There is no restriction $n \geq 2$ in either the English or Chinese statement. The polynomial assertion is interpreted in its substantive, universal sense: for each n, one polynomial works for all graphs. Allowing a different polynomial for each individual graph would impose no relation between these invariants.

Throughout, a finite graph denotes its usual one-dimensional CW realization, not its discrete vertex set. Its first Betti number is $\beta_1(G) = \dim_{\mathbb{Q}} H_1(G;\mathbb{Q})$. Both graphs below are finite, simple, nonempty, and connected.

Theorem 1. *There is no function of chromatic number alone that equals $\beta_1(\mathrm{Conf}_1(G))$ for every finite simple connected graph G. Consequently the polynomial clause of conjecture 00000004091, and hence its conjunction with subdivision invariance, is false.*

2 The one-particle reduction

For a topological space X, the ordered configuration space is

$$F_n(X) = \{(x_1, \ldots, x_n) \in X^n : x_i \neq x_j \text{ for } i \neq j\}, \qquad \mathrm{Conf}_n(X) = F_n(X)/S_n.$$

This is the standard definition used for graphs [3, p. 2]. When $n = 1$, the distinctness requirement has no instances, so $F_1(X) = X^1$. Evaluation $(x) \mapsto x$ and insertion $x \mapsto (x)$ are continuous inverse maps for the product topology. The group S_1 has one element, so every orbit is a singleton. The quotient map $q : X^1 \to X^1/S_1$ is bijective; it is a homeomorphism because the quotient-topology definition says that U in the quotient is open exactly when $q^{-1}(U)$ is open. Thus the map $[(x)] \mapsto x$ is a natural homeomorphism

$$\mathrm{Conf}_1(X) \cong X.$$

Homeomorphism invariance of homology reduces the theorem to the two graph calculations below.

3 Exact graph calculations

3.1 Chromatic numbers

The graph K_2 has vertices $0, 1$ and its single edge joins them. Coloring these vertices $0, 1$ gives a proper two-coloring. Since the edge endpoints must have different colors, one color cannot suffice. Hence $\chi(K_2) = 2$.

Let C_4 have vertices $0, 1, 2, 3$ and edges $\{0, 1\}, \{1, 2\}, \{2, 3\}, \{3, 0\}$. The parity coloring $i \mapsto i \bmod 2$ is proper. The presence of an edge again excludes one color. Thus $\chi(C_4) = 2$.

3.2 Rational cellular homology

Choose one orientation for every edge. The cellular chain spaces are $C_1(G; \mathbb{Q}) = \mathbb{Q}^{E(G)}$, $C_0(G; \mathbb{Q}) = \mathbb{Q}^{V(G)}$, and $C_2(G; \mathbb{Q}) = 0$. For an edge e directed from u to v, the boundary is $\partial_1 e = v - u$. Therefore

$$H_1^{\text{cell}}(G; \mathbb{Q}) = \ker \partial_1 / \operatorname{im} \partial_2 = \ker \partial_1.$$

This is the standard graph chain description in Hatcher [2, pp. 99–101]. Cellular homology agrees with singular homology by Theorem 2.35, and the same comparison with arbitrary coefficients is explained in the section beginning on p. 153 [2].

For K_2, orient its edge $0 \to 1$. The boundary matrix is

$$B_{K_2} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

A chain with coefficient a has zero boundary precisely when $(-a, a) = (0, 0)$, which forces $a = 0$. Thus the kernel has its empty basis and $\beta_1(K_2) = 0$.

For C_4, use the coherent orientations $0 \to 1 \to 2 \to 3 \to 0$. In this order the boundary matrix is

$$B_{C_4} = \begin{pmatrix} -1 & 0 & 0 & 1 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}.$$

The kernel equations for $z = (a, b, c, d)$ are $d - a = 0$, $a - b = 0$, $b - c = 0$, and $c - d = 0$. They are equivalent to $a = b = c = d$. Hence every cycle has the unique expression $a(1, 1, 1, 1)$, and the nonzero all-one chain is a basis. It follows that $\beta_1(C_4) = 1$.

3.3 Contradiction

If one function f of chromatic number supplied the claimed Betti number, these two calculations and the one-particle homeomorphism would give

$$f(2) = 0 \qquad \text{and} \qquad f(2) = 1,$$

which is impossible. In particular, no polynomial of degree exactly 1 can do so for $n = 1$. No assertion about degree or polynomial coefficients is needed for this contradiction. This proves the theorem.

Subdivision invariance itself is compatible with this argument: subdivision does not change the topological realization of the graph, and the induced homeomorphism acts on configuration spaces coordinatewise. It is the chromatic-determination clause that fails.

4 Lean verification and its scope

The project imports only `Std` and targets Lean 4.31.0. Its finite graph structure stores oriented endpoints and proves that there are no loops or repeated undirected edges. The source `Conjecture4091/Basic.lean` contains:

- `boundary`: the actual incoming-minus-outgoing incidence map, with edge coefficients in Lean's exact rational type `Rat`.
- `IsCycle`: the kernel equations at every vertex.
- `HasFirstBetti G d`: an actual d-vector basis of that kernel, expressed by spanning and uniqueness of rational linear-combination coefficients. Betti values are not assigned as labels.
- `segment_cycle_iff` and `square_cycle_iff`: complete characterizations of the two incidence kernels, followed by basis certificates `segment_firstBetti` and `square_firstBetti`.
- `HasChromaticNumber`: a legal coloring and a proof that every smaller color count is impossible. Both witnesses are certified at 2.
- `conjecture_00000004091_false`: negation of the necessary one-particle consequence that an arbitrary function $\mathbb{N} \to \mathbb{N}$ determines the first Betti number from chromatic number. The contradiction uses the two certificates at the same input 2.

4.1 Explicit bridge to the topological statement

For a stored graph G, form $|G|$ by taking one point for each stored vertex and one copy of $[0, 1]$ for each stored edge, identifying 0 with its source and 1 with its target. This is the finite one-dimensional CW realization with exactly the stored vertices and open edges as cells. The source and target data of `segment` give K_2; those of `square` give the four edges of C_4 listed above.

Sending a rational function on the stored edges to the corresponding linear combination of oriented cellular edges is a vector-space isomorphism $\mathbb{Q}^{\mathrm{Fin}(G.\mathrm{edges})} \to C_1(|G|; \mathbb{Q})$; the analogous map identifies stored vertex functions with C_0. Under these maps, the coefficient of a vertex in the cellular boundary is exactly the sum on incoming edges minus the sum on outgoing edges. Thus Lean's `boundary` is the cellular boundary, not an assigned numerical invariant. Its `linearCombination` is the ordinary rational linear combination in these coordinates. Lean's `Rat` represents the field of rational numbers exactly.

A witness for `HasFirstBetti G d` is therefore a genuine basis of $\ker \partial_1$: every proposed vector lies in the kernel, every kernel vector is spanned, and the coefficients are unique. Uniqueness implies linear independence by applying it to the zero vector. Consequently its d equals $\dim_{\mathbb{Q}} \ker \partial_1$. Because $|G|$ has no cells in dimension two, $\operatorname{im} \partial_2 = 0$. The cellular-to-singular comparison cited above now identifies this kernel with $H_1(|G|; \mathbb{Q})$. Finally, the homeomorphism from Section 2 identifies it with $H_1(\mathrm{Conf}_1(|G|); \mathbb{Q})$. For colorings, choosing an orientation does not affect whether the two endpoints have different colors, so `HasChromaticNumber` gives the usual chromatic minimum of the underlying undirected graph. These identifications verify the semantic link for each witness.

The formal dimension certificates use the standard cellular definition of first Betti number for a graph. The quotient topology of Conf_1, its homeomorphism with G, and the general cellular-to-singular comparison are mathematical bridges proved or cited above; they are not machine-checked by this Std-only project. No such bridge is introduced as a Lean axiom. The

proof is a semantic counterexample in the graph cellular model, with this topological boundary stated explicitly.

The final theorem and basis theorems use only the standard Lean logical axioms `propext`, `Classical.choice`, and `Quot.sound`. There are no `sorry` placeholders, `native_decide`, or custom axioms. The included build logs record successful compilation and a separate `leanchecker` run.

5 Independent computation and eligibility

The separate `empirical.py` script performs exact Gaussian elimination over Python rational fractions and exhaustive coloring searches. It confirms the incidence ranks 1 and 3 and kernel dimensions 0 and 1 of the two witnesses. It also enumerates all 1,099 labeled simple graphs with one through five vertices. Among connected bipartite graphs in this range, it finds first Betti numbers $0, 1, 2$ at the same chromatic number 2. These computations are independent corroboration; they are not premises of the Lean proof.

At the live check on 14 October 2026 at 19:46 UTC, conjecture 00000004091 had `proven=false` and `disproven=false`; its solutions path was absent. All-state searches for the full ID and for Betti/subdivision found the earlier, closed, unmerged submission PR 292 [4]. The short-ID and configuration/chromatic searches returned no matches; these inconsistent search results are why the prior PR was also fetched directly. The response snapshot is retained in `evidence/eligibility-00000004091.json`. This records a check at that time, not acceptance or priority over earlier or later work.

6 Earlier submission and the present repair

The counterexample idea is already present in orionsheep's PR 292 of 3 October 2026: at one particle, a path and an even cycle have the same chromatic number but different first Betti numbers. We credit that earlier observation and do not claim it as a new mathematical discovery. That PR used P_4 and C_4; the present proof uses the shorter path K_2 and the same cycle.

The organizer closed PR 292 without merging it. The recorded reason was that its Lean statement only conjoined elementary integer equalities and inequalities, with no graph, configuration-space, or Betti-number objects connecting the arithmetic to the conjecture [5]. Inspection of that PR's source confirms the issue: `b1_C4` and `b1_P4` are just arithmetic identities, `same_chromatic` is the conjunction $2 = 2 \wedge 2 = 2$, and `conjecture_refuted` merely assembles those statements. This is a formalization gap; the elementary paper counterexample is sound.

The present project supplies the missing finite simple graphs, proper colorings and minimality proofs, incidence boundary on rational chains, and kernel bases with unique coefficients. Its concluding theorem refutes a universally quantified dependence on chromatic number using those certificates. The exact bridges from this cellular model to the topological conjecture are given in Section 4.1. They remain paper mathematics, rather than Lean theorems. Thus this submission repairs the graph/chromatic/Betti semantic gap but does not claim a full Lean formalization of general configuration-space topology. Whether this disclosed boundary meets the competition's review standard remains for its reviewers to decide. A detailed comparison is included in `PRIOR_SUBMISSION.md`.

References

[1] The Last Math Competition, conjecture 00000004091. https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/blob/main/conjectures/000000004091.md. Source blob: 73f51cfa010d743d7f2363be23ee139f9c819802.

[2] Allen Hatcher, *Algebraic Topology*, Cambridge University Press, 2002. Chapter 2, pp. 99–101; Theorem 2.35, pp. 139–140; homology with coefficients, p. 153. https://pi.math.cornell.edu/~hatcher/AT/ATch2.pdf.

[3] Steven Scheirer, *Topological complexity of unordered configuration spaces of certain graphs*, 2019, p. 2. https://arxiv.org/pdf/1905.12838.

[4] orionsheep, PR 292, *Disproof of conjecture 00000004091*, 3 October 2026, closed without merging. https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/pull/292.

[5] lidangzzz, review comment on PR 292, 3 October 2026. https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/pull/292#issuecomment-5968931449.